

Dok. 261 — Static Geometry and the SI Projection: Why Natural Units Require No Fractal Corrections

Johann Pascher

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units), Dok. 116 (Koide), Dok. 133 (fractal correction)
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1 Setting

The 2019 SI reform fixed four constants (c , $\hbar$, e , k_B) to exact values and thereby turned most dimensionful quantities into mere unit definitions. What remains as *genuine* empirical input are the dimensionless constants – above all the fine-structure constant α , whose value is still measured after the reform (μ_0 , ε_0 and α remained measured quantities). FFGFT derives these dimensionless inputs as well, from the single parameter $\xi = \frac{4}{3} \times 10^{-4}$.

This document collects the conversion machinery compactly (Tables 1–3) and then establishes one central point: *as long as one stays in natural units and pure ratios, the description is static and relative and requires no fractal corrections*. The fractal corrections appear only upon projection to absolute SI values.

2 The constants fixed by the SI reform

Table 1: The constants fixed by the 2019 SI reform (the “machinery”). They carry no physics, only scale, and become 1 in natural units.

Constant	Value (exact since 2019)	defines	natural units
c	299 792 458 m/s	metre	= 1
$\hbar$	$1.054 571 817 \dots \times 10^{-34}$ J s	kg	= 1
e	$1.602 176 634 \times 10^{-19}$ C	ampere	charge unit
k_B	$1.380 649 \times 10^{-23}$ J/K	kelvin	= 1

Table 2: Quantity conversion in natural units ($\hbar = c = k_B = 1$): all dimensionful quantities become powers of energy.

Quantity	SI dimension	$[\hbar = c = k_B = 1]$	back to SI
Energy	J	E	base
Mass	kg	E	$\times S_{T0}$ (via c^{-2})
Momentum	kg m/s	E	$\times c^{-1}$
Length	m	E^{-1}	$\ell[\text{fm}] = 197.327/E[\text{MeV}]$
Time	s	E^{-1}	$t[\text{s}] = 6.582 \times 10^{-22}/E[\text{MeV}]$
Temperature	K	E	$T[\text{K}] = E[\text{eV}]/(8.617 \times 10^{-5})$
Charge (Gauss)	C	dimensionless	$e = \sqrt{\alpha} \approx 0.0854$

The conversion to SI runs through the *single* mass scale

$S_{T0} = 1 \text{ MeV}/c^2 = 1.782662 \times 10^{-30} \text{ kg}$ (Dok. 013/014); the length factor is $\hbar c/S_{T0} = 1.973 \times 10^{-13} \text{ m}$, i.e. the 197 MeV fm of the $\hbar c = 1$ column expressed in T0 language.

In T0 units the charge unit is chosen ($e = \sqrt{4\pi\epsilon_0\hbar c}$) so that the *written* value is $\alpha = 1$ (Dok. 044) – a convention, not a physical intervention. The physical invariant $\alpha \approx 1/137$ is reconstructed from ξ (Table 3).

Table 3: The surviving dimensionless inputs of physics and their derivation from the single parameter $\xi = \frac{4}{3} \times 10^{-4}$. All relations are sourced.

Invariant	Standard physics	FFGFT relation (from ξ)	Source
ξ	—	$= \frac{4}{3} \times 10^{-4} = 4/30000$	Dok. 013
α (SI)	$1/137.036$	$\xi (E_0/1 \text{ MeV})^2$ [†]	Dok. 011
Lepton masses	measured	$m = r \xi^p v$	Dok. 116, 006
exponents		$(p_e, p_\mu, p_\tau) = (\frac{3}{2}, 1, \frac{2}{3})$	Dok. 116
ratios		$(r_e, r_\mu, r_\tau) = (\frac{4}{3}, \frac{16}{5}, \frac{25}{9})$	Dok. 116
Koide Q	0.66666	$= \frac{2}{3}$ (consequence of the ξ exponents)	Dok. 116, 258
$\sin^2 \theta_W$	0.2312	$\frac{1 - \sqrt{1 - 4\alpha_W}}{4}, \alpha_W = \sqrt{\xi}$	Dok. 041
θ_{13}	8.57°	$\arcsin(\xi^{1/3})$	Dok. 041
G	measured	$\frac{\xi^2}{4m_e} K_{\text{frak}}, K_{\text{frak}} = 1 - 100\xi \approx 0.9867$	Dok. 012, 133

[†] $\alpha = \xi (E_0/1 \text{ MeV})^2$ is the *leading order*, with $E_0 = \sqrt{m_e m_\mu} = 7.398 \text{ MeV}$.

CODATA precision is reached through the fractal correction product

$\prod_{n=1}^{137} (1 + \delta_n \xi (4/3)^{n-1})$. These corrections are **not free assumptions**:

$K_{\text{frak}} = 1 - 100\xi$ is uniquely fixed by the scale flow from the Planck to the electroweak scale ($D_f^{\text{space}} = 3 - \xi$, Dok. 133, “ D_f is uniquely determined – not freely choosable”), and the rational ladder ratios r_i are α -geometric invariants (Dok. 258), structurally inaccessible to any after-the-fact fitting (non-closure theorem, Dok. 193).

3 Quantity conversion in natural units

4 The dimensionless inputs and their derivation from ξ

5 Static geometry: why natural units are correction-free

As long as physics is expressed in natural units and pure ratios, its description is *static* and *relational*: only dimensionless ratios occur, no absolute anchors and no scale dependence. At this level the FFGFT quantities are exact geometric invariants of ξ – and no fractal correction is required.

Principle. In natural units FFGFT is purely relational. The dimensionless structural relations – mass ratios, mixing angles, the Koide quantity – are static geometric invariants of ξ . They contain no fractal corrections.

The clearest example is the Koide quantity: $Q = \frac{2}{3}$ follows exactly from the exponents $(p_e, p_\mu, p_\tau) = (\frac{3}{2}, 1, \frac{2}{3})$ and the ratios $(r_e, r_\mu, r_\tau) = (\frac{4}{3}, \frac{16}{5}, \frac{25}{9})$ – with no additional parameter and no fractal correction (Dok. 116). Likewise the ladder ratios r_i , the exponents p_i , the Weinberg structure and θ_{13} are exact functions of ξ alone. None of these relations needs an absolute scale, an E_0 in MeV, or a conversion factor.

Where, then, do the fractal corrections arise? Only at the *SI projection* – where an absolute scale is fixed. $K_{\text{frak}} = 1 - 100\xi$ is not a correction to the geometry but the cumulative effect of the scale flow from the Planck to the electroweak scale (Dok. 133): a property of mapping the geometry onto an absolute energy ruler. Likewise the fractal correction product $\prod_n (1 + \delta_n \xi (4/3)^{n-1})$ appears only when the leading relation $\alpha = \xi(E_0/1 \text{ MeV})^2$ is brought to CODATA precision in absolute SI values.

An important delimitation: these projection factors are nonetheless *not* free parameters. K_{frak} is uniquely fixed by the scale flow (Dok. 133), the rational ratios are α -geometric invariants (Dok. 258), and the non-closure theorem (Dok. 193) blocks any after-the-fact adjustment. They are derived projection factors, not assumed fits.

Conclusion. The fractal corrections are the price of the absolute projection, not a defect of the geometry. As long as one stays in natural units and ratios, the description is static, relative and correction-free; the corrections appear exclusively at the transition to absolute SI values – and even there they are uniquely determined by ξ and the scale structure.